Certified Robustness from One Setpoint Step: A Safe Acceptance Test for Model-Free PID Tuning

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Abstract

Model-free PID tuners improve a loop from closed-loop data, but they do not guarantee its robustness, measured by the peak M_s of the sensitivity function. We show that the setpoint step such tuners already apply certifies M_s: in a stable digital loop, the error increments after a unit setpoint step are the impulse response of the sensitivity function, so a finite record gives computable lower and upper bounds on M_s, under one assumption on the unrecorded rest of the response. The same record, from a stable running loop, certifies the stability and bounds the M_s of any other controller before it is applied. Any iterative tuner can thus accept only tunings with $M_s \le M^*$ and never apply one above M^*. For SPIN, which tunes by counting the turns of the control error in phase portraits, the counts bound M_s only on special loops. We add the certificates in two steps: certified SPIN checks each tuning after applying it, safe SPIN before. On 19 simulated loops, published SPIN ended above $M^* = 1.7$ on up to 9 and applied tunings above it in 57% of its iterations; certified SPIN ended below M^* on all loops but still applied such tunings in 45%; safe SPIN never applied one.

1 Introduction

Let C be the controller, P the plant, $L = CP$ the loop transfer function, $S = 1/(1+L)$ the sensitivity function and $T = 1 - S$ the complementary sensitivity function. The peak $M_s = \max_\omega |S|$ measures robustness: $1/M_s$ is the shortest distance from the Nyquist curve of L to the critical point -1, so M_s near 1.4 is safe and near 2 is poor [1]. Model-based tuning rules reach a chosen M_s [1, 2]. Model-free tuners adjust the gains from closed-loop data instead, by iterative feedback tuning [3], extremum seeking [4], virtual reference feedback tuning [5], setpoint overshoot [6] or step-response inspection [7]. They target performance or a response shape; none guarantees the M_s of its result, or of the tunings it tries on the way. Once a loop runs, its M_s is rarely checked at all [8, 9].

The $\mathscr{H}_\infty$ norm of a system, and hence M_s, can be estimated from data: by power iterations over repeated dedicated experiments [10, 11], by data-driven estimation [12], from limit cycles induced in closed loop [13], or from impulse-response data under a decay assumption, with a one-sided guarantee [14]; margins can be bounded from data [15] or measured by relay tests and closed-loop identification [16, 17]. These methods need dedicated experiments, give estimates or one-sided bounds, and are not built into a tuner. Safe Bayesian optimization tunes controllers without leaving a safe set [18], with probabilistic guarantees under a Gaussian-process model of the cost.

We use the experiment a tuner already performs: a setpoint step. That the error increments after the step are the impulse response of S is elementary (Lemma 1). The contributions are what it enables, for a chosen target M^*:

1. a deterministic, two-sided certificate of M_s from one finite record (Section 2, Theorem 1);
2. for any iterative tuner, a certified acceptance test (Theorem 2) and a certificate of the stability and M_s of a tuning *before* it is applied (Theorem 3), which keeps the loop robust during the tuning (Section 3);
3. their application to SPIN [7]: why its turn counts cannot certify M_s (Lemma 2, Proposition 1), certified SPIN, which ends below M^*, and safe SPIN, which never applies a tuning above it (Section 4, Corollaries 1 and 2).

Section 5 confirms the results in simulation.

2 A Certificate from One Setpoint Step

A digital controller runs with the sampling period T_s. Seen at the sampling instants, the loop it closes is linear and time invariant, with transfer functions $C(z)$, $P(z)$, L, S and T as above. On the unit circle, $z = \mathrm{e}^{j\theta}$ with the normalized frequency $\theta = \omega T_s \in [0, \pi]$, and $M_s = \max_\theta |S(\mathrm{e}^{j\theta})|$; we write $M_s(C)$ when the controller matters. All these quantities refer to the digital loop, the loop that actually runs; e (upright) is Euler's number.

Lemma 1 (The step contains S) *Let the loop be stable and have one degree of freedom (the controller acts on the error*

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$e = r - y$, with setpoint r and output y), and let L be strictly proper (the measurement precedes the control action). Let e_n be the error after a unit setpoint step at $n = 0$, applied from rest and without disturbances, so that $e_{-1} = 0$ and $e_0 = 1$, and let $a_n = e_n - e_{n-1}$. Then

$$S(\mathrm{e}^{j\theta}) = \sum_{n\geq 0} a_n \mathrm{e}^{-jn\theta}. \tag{1}$$

Proof: Let $R(z)$ and $E(z)$ be the z-transforms of the setpoint and the error. A unit step has $R = 1/(1-z^{-1})$, and $E = SR$, so $S = (1-z^{-1})E$, whose coefficients are the increments a_n. In a stable loop, a_n decays exponentially, so the series converges on the unit circle. □

A real record $e_0, \ldots, e_N$ is finite, so one assumption covers the increments it does not show.

Assumption 1 (Long enough record) *The loop is stable, and the record $e_0, \ldots, e_N$ satisfies $\sum_{n>N} |a_n| \leq \tau$ for a known τ.*

A valid τ falls geometrically with the record length (Section 5 checks it with a longer record). No finite record can show that the loop is stable or the tail small: an unstable or slow mode may appear only after it ends. Theorem 1 thus bounds M_s of a loop assumed stable; Theorem 3 then certifies the stability of each next controller from the record of the current one. By (1), the record gives the estimate $\hat{S}(\theta) = \sum_{n=0}^{N} a_n \mathrm{e}^{-jn\theta}$ of S at every frequency. Three errors separate the largest $|\hat{S}|$ on a frequency grid from M_s: the unrecorded tail, the gaps between grid points, and high frequencies, which a grid resolves only at great cost. Theorem 1 bounds each of them.

Theorem 1 (Certified bounds from one record) *Let Lemma 1 and Assumption 1 hold. Let G be a uniform grid on $[0, \bar{\theta}]$, $0 < \bar{\theta} \leq \pi$, with spacing h and containing both ends. For any integer n_c, let*

$$J = \textstyle\sum_{n=0}^{N} |n - n_c|\,|a_n|, \quad B = \sum_{n=1}^{N-1} |a_{n+1} - a_n| + |a_N| + 2\tau.$$

Then $\underline{M} \leq M_s \leq \overline{M}$, where

$$\underline{M} = \max_{\theta\in G} |\hat{S}(\theta)| - \tau,$$
$$\overline{M} = \max\left\{ \max_{\theta\in G} |\hat{S}(\theta)| + \tfrac{hJ}{2} + \tau,\ 1 + \tfrac{B}{\sin(\bar{\theta}/2)} \right\}, \tag{2}$$

and the second term is omitted if $\bar{\theta} = \pi$. If τ equals the true remaining variation $\sum_{n>N} |a_n|$, then both bounds converge to M_s as $N \to \infty$ and $h \to 0$ with $\bar{\theta} = \pi$.

Proof: Tail. By (1), $S - \hat{S} = \sum_{n>N} a_n \mathrm{e}^{-jn\theta}$, so $|S - \hat{S}| \leq \tau$ at every θ. This gives $\underline{M}$ and the term τ in $\overline{M}$. *Grid.* On $[0, \bar{\theta}]$, $|\hat{S}(\theta)| = |\sum_n a_n \mathrm{e}^{-j(n-n_c)\theta}|$, whose derivative with respect to θ is at most J in magnitude. Every θ lies within $h/2$ of a grid point, so $|\hat{S}|$ exceeds its largest grid value by at most $hJ/2$. *High frequencies.* On $[\bar{\theta}, \pi]$, let $z = \mathrm{e}^{-j\theta}$ and $Z_n = \sum_{m=1}^{n} z^m$, so that $|Z_n| = |z(1-z^n)/(1-z)| \leq 2/|1-z| = 1/\sin(\theta/2)$.

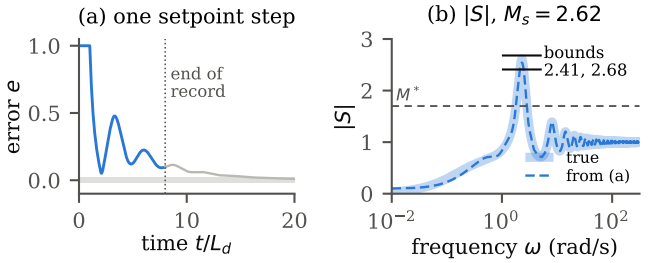


Figure 1: One step certifies M_s: a first-order plant with dead time L_d and time constant $T_1 = 0.3L_d$ under PI with integral time $T_i = 8T_1$, $T_s = 0.01L_d$. (a) The error after a setpoint step; it never crosses the setpoint by more than 2% (grey band). The record ends at $8L_d$. (b) $|\hat{S}|$ from the record lies on the true $|S|$, and the bounds (2) bracket $M_s = 2.62$, far above M^* ($\tau = 0.13$, the true variation after $8L_d$).

Summation by parts gives $\sum_{n\geq 1} a_n z^n = \sum_{n\geq 1} (a_n - a_{n+1}) Z_n$, and $\sum_{n\geq N} |a_n - a_{n+1}| \leq |a_N| + 2\sum_{n>N} |a_n| \leq |a_N| + 2\tau$. So $|S - a_0| \leq B/\sin(\theta/2) \leq B/\sin(\bar{\theta}/2)$ with $a_0 = 1$. *Convergence.* For a fixed n_c, J stays bounded as $N \to \infty$, so $hJ \to 0$, and the remaining variation tends to zero. □

Everything in (2) except τ comes from the record; n_c only affects tightness (we take the weighted median of $|a_n|$). All grid values $\hat{S}$ follow from one chirp z-transform, an FFT-type algorithm. The grid term needs a fine grid where $|\hat{S}|$ changes fast; the high-frequency term needs no grid at all, and B is small for a smooth record. Fig. 1 shows both bounds on one loop.

3 A Certified Acceptance Test

From here on, the plant is stable (so $P(1)$ is finite), as are the self-regulating processes that SPIN targets; Lemma 1 and Theorem 1 need only a stable closed loop. An iterative tuner applies a controller C, records the error after a setpoint step, and proposes the next controller from the record. We call a set of gains a *tuning*, and say that a tuning is *accepted* when the tuner keeps it as a candidate result. *Certified acceptance* keeps the tuner's own rule for proposing tunings, but accepts C only if the upper bound $\overline{M}$ of Theorem 1, computed from the record of C, is at most the target M^*. Otherwise it lowers the gains, by the tuner's own rule or by a *uniform back-off* $C \to (1-\beta)C$ with $0 < \beta < 1$.

We assume integral action, $C = C_0 + K_i T_s/(z-1)$, where the integral gain is $K_i > 0$ and C_0, the proportional and derivative part, is stable. In (2), we choose $\bar{\theta}$ so that the high-frequency term equals M^*, that is, $1 + B/\sin(\bar{\theta}/2) = M^*$ ($\bar{\theta} = \pi$ if $1 + B > M^*$), and the grid so fine that $hJ/2 \leq \eta$ for a small tolerance η.

Theorem 2 (Certified acceptance) *Let the plant be stable, let $K_i P(1) > 0$ (the integral action counteracts the error), and let every record meet Lemma 1 and Assumption 1.*

(i) Every accepted tuning has $M_s \le M^*$, whatever rule proposed it.

(ii) For a controller C, let $\mu = \min\{0, \inf_{0<\theta\le\pi} \mathrm{Re}\, L(e^{j\theta})\}$, the leftmost real part of its Nyquist curve. For $0 < \lambda < 1/|\mu|$, the controller λC stabilizes the plant, and

$$M_s(\lambda C) \le \frac{1}{1-\lambda|\mu|}. \tag{3}$$

(iii) If $M^* > 1 + 2\tau + \eta$, then at most $j^* = \lceil \ln \frac{|\mu|M'}{M'-1} / \ln \frac{1}{1-\beta} \rceil$ uniform back-offs from any such C lead to an accepted tuning, where $M' = M^* - 2\tau - \eta$.

Proof: (i) Accepted means $\overline{M} \le M^*$, and $M_s \le \overline{M}$ by Theorem 1. (ii) First, μ is finite: L is bounded on the unit circle except near $z = 1$, where the integral term has the real part $-K_i T_s/2$, since $\mathrm{Re}\, 1/(e^{j\theta} - 1) = -\frac{1}{2}$, and P is continuous at $z = 1$. For every θ, $|1 + \lambda L| \ge 1 + \lambda\, \mathrm{Re}\, L \ge 1 - \lambda|\mu| > 0$, which gives (3). For stability, we apply the Nyquist criterion on the unit circle with a small indentation that passes outside the pole $z = 1$. On the indentation, $\lambda L \approx \lambda K_i T_s P(1)/(z-1)$ has a positive real part, since $K_i P(1) > 0$. So the whole Nyquist curve of λL lies in $\mathrm{Re} \ge -\lambda|\mu| > -1$ and cannot encircle -1; as λL has no poles outside the indented circle, the loop is stable. (iii) After j back-offs, $\lambda = (1-\beta)^j$, and (3) gives $M_s \le M'$ once $j \ge j^*$. Then the grid term of (2) is at most $(M_s + \tau) + \eta + \tau \le M^*$, and the high-frequency term is at most M^* by the choice of $\bar{\theta}$; so the tuning is accepted. □

Acceptance tests a tuning after it has been applied: the tuner learns that a tuning is fragile only by running it. The record of C, however, determines the sensitivity of any other controller C' on the same plant, so a tuner can check C' before applying it. Let $\rho = C'/C - 1$ be the relative change of the controller. Then $L' = L(1+\rho)$, so $1 + L' = 1 + L + L\rho = (1+L)(1+T\rho)$, and

$$S' = \frac{S}{1+T\rho}. \tag{4}$$

The record gives $\hat{S}$ and $\hat{T} = 1 - \hat{S}$, each within τ.

Theorem 3 (Certify before applying) *On a stable plant, let C and C' be controllers as in Theorem 2 with the same T_s, let ρ be bounded on the unit circle, and let the record of C meet Lemma 1 and Assumption 1. If, for all $\theta \in (0, \pi]$,*

$$d(\theta) = \min_{0\le\lambda\le 1} |1 + \lambda\hat{T}\rho| - \tau|\rho| > 0, \tag{5}$$

then C' stabilizes the plant, and

$$M_s(C') \le \sup_{0<\theta\le\pi} \frac{|\hat{S}| + \tau}{|1 + \hat{T}\rho| - \tau|\rho|}. \tag{6}$$

The minimum in (5) is the distance from 0 to the segment between 1 and $1 + \hat{T}\rho$.

Proof: The controllers $C_\lambda = C(1 + \lambda\rho) = (1-\lambda)C + \lambda C'$ move from C to C' as λ goes from 0 to 1. Each has a positive integral gain, and its other poles are those of C_0 and C_0', inside the unit disc. So $L_\lambda = C_\lambda P$ has no poles outside the indented circle of Theorem 2, and on the indentation it has a large positive real part. By (4) with $\lambda\rho$ in place of ρ, $1 + L_\lambda = (1+L)(1+\lambda T\rho)$. Here $1 + L \ne 0$, and since $|T - \hat{T}| = |S - \hat{S}| \le \tau$, $|1 + \lambda T\rho| \ge |1 + \lambda\hat{T}\rho| - \lambda\tau|\rho| \ge d > 0$. So $1 + L_\lambda \ne 0$ on the indented circle for every λ, and the number of times its Nyquist curve encircles 0 cannot change with λ. The loop is stable at $\lambda = 0$, hence at $\lambda = 1$. The bound (6) follows from (4), $|S| \le |\hat{S}| + \tau$ and $|1 + T\rho| \ge |1 + \hat{T}\rho| - \tau|\rho|$. □

In practice, (5) and (6) are checked on the grid of Theorem 1, with margins that bound how much each term can change between grid points. Above $\bar{\theta}$, the bound $|S - 1| \le r = B/\sin(\theta/2)$ from the proof of Theorem 1 replaces $\hat{S}$: then $|T| \le r$ and $|S| \le 1 + r$, so (5) holds if $r|\rho| < 1$, and $|S'| \le (1+r)/(1 - r|\rho|)$. Since (4) is exact, Theorem 3 needs no small-step assumption; a larger change only widens the bound through $\tau|\rho|$. As $\tau \to 0$, (6) becomes $\max_\theta |\hat{S}/(1 + \hat{T}\rho)|$, which we call the *prediction* of $M_s(C')$. A uniform back-off is the case $\rho = -\beta$.

4 Application to SPIN

4.1 The SPIN method

SPIN [7] tunes an I, PI or PID controller with the integral, proportional and derivative gains K_i, K_p, K_d. In each iteration, it applies one setpoint step, records the error e_n, and forms three phase portraits, curves in the plane traced as n grows:

$$\begin{aligned} &\Gamma_0 : \big(e_n, \textstyle\sum_{m\le n} e_m\big), \qquad \Gamma_1 : (\Delta e_n, e_n), \\ &\Gamma_2 : (\Delta^2 e_n, \Delta e_n), \end{aligned} \tag{7}$$

with the differences $\Delta e_n = e_n - e_{n-1} = a_n$ and $\Delta^2 e_n = \Delta e_n - \Delta e_{n-1}$. Γ_0, Γ_1 and Γ_2 emphasize low, middle and high frequencies, where the integral, proportional and derivative actions dominate the controller; we call them bands 0, 1 and 2. Each axis is divided by its peak, and the count N_k is the number of turns that Γ_k makes around its end point until the error stays within $\pm\delta$ of the setpoint (δ is 2% of the step). Algorithm 1 states how the counts change the gains.[1]

4.2 Why the turn counts cannot certify

Write a portrait (7) as (p, q), with q the running sum of p, shifted so that the end point is the origin. The count is the net counterclockwise angle swept around the origin, divided by 2π. We state the results in continuous time, where $\dot{q} = p$. For Γ_0, $p = e$, and its zeros are the setpoint crossings.

Lemma 2 (Half turns) *Let $\dot{q} = p$ with p continuous, let $p(t_1) = p(t_2) = 0$ with $q(t_1)q(t_2) < 0$, and let $p \ne 0$ on*

[1] An online simulator: `https://robopid-simulator.uceeb.cvut.cz`

Algorithm 1: one iteration of SPIN [7] and, with lines 7a–7c in bold instead of line 7, of certified SPIN (Section 4.3)

```
 1: apply a setpoint step and record the error e
 2: if the record does not settle then
 3:     divide K_i, K_p, K_d by 2, 4, 8                (retreat)
 4: else
 5:     count the turns N_0, N_1, N_2 of Γ_0, Γ_1, Γ_2
 6:     if N_k ≤ N̄_k for every band k then
 7:         raise all gains by one increment           (accept)
7a:         compute M̄ from the record (Theorem 1)
7b:         if M̄ ≤ M* then raise all gains             (accept)
7c:         else lower the gains by a band move
               (line 10) or a uniform back-off         (reject)
 8:     else
 9:         k ← lowest band with N_k > N̄_k
10:         lower the gain of band k by one increment and
            raise the gains of the bands below k by one
```

An increment scales a gain by $1/(1-\beta)$. Defaults: limits $\bar{N}_k = (0.5, 0.75, 1.0)$, $\delta = 0.02$, $\beta = 0.1$; gains stay within $[0.001, 10]$ times their initial values (the gain box); the result is the last accepted tuning. Certified SPIN: $M^* = 1.7$, $\tau = 10^{-3}$, $\eta = 2 \cdot 10^{-3}$, $\bar{\theta}$ and h as in Section 3.

(t_1, t_2). *Then, on* $[t_1, t_2]$, *the curve* (p, q) *turns counterclockwise around the origin by exactly half a turn, and so does its image under any linear map with a positive determinant, such as the per-axis scaling of SPIN.*

Proof: On (t_1, t_2), p keeps one sign, so the curve stays in one half-plane bounded by the q-axis, and q moves monotonically from one ray of the q-axis to the opposite ray. The curve avoids the origin, since $p = 0$ only at t_1 and t_2, where $q \neq 0$. The angle thus changes by exactly π, counterclockwise, since q increases where $p > 0$ and decreases where $p < 0$. A linear map with a positive determinant maps half-planes to half-planes and opposite rays to opposite rays, and keeps the orientation. □

So each setpoint crossing adds half a turn to N_0, and a count sees the crossings that rise above δ. Whether crossings reveal M_s depends on the loop. We first study a loop where they do. Lambda, IMC and SIMC tuning rules cancel the plant lag by the controller zero [2], e.g., PI with the integral time $T_i = T_1$ on the plant $K\mathrm{e}^{-sL_d}/(T_1 s + 1)$. The loop is then

$$L(s) = \frac{\kappa\, \mathrm{e}^{-sL_d}}{sL_d}, \qquad \kappa = \frac{KK_pL_d}{T_i}. \tag{8}$$

Proposition 1 (Crossings on lag-cancelled loops) *The loop* (8) *is stable for* $0 < \kappa < \pi/2$. *Its* M_s *strictly increases with* κ *and is given by*

$$\kappa = x\sin x - x^2\cos x, \qquad \frac{1}{M_s^2} = 1 - \frac{2\kappa\sin x}{x} + \frac{\kappa^2}{x^2}, \tag{9}$$

where $x \in (0, \pi/2)$ *is the peak frequency normalized as* $x = \omega L_d$. *The error after a unit setpoint step never crosses the setpoint if and only if* $\kappa \leq 1/\mathrm{e}$, *that is,* $M_s \leq 1.394$; *otherwise it crosses infinitely often.*

Proof: At $x = \omega L_d$, L has the phase $-\pi/2 - x$ and the gain κ/x. The phase first reaches $-\pi$ at $x = \pi/2$, where the gain is $2\kappa/\pi < 1$, so the loop is stable. Let $f(x) = |1 + L|^2 = 1 - 2\kappa\sin x/x + \kappa^2/x^2$, so that $M_s^2 = 1/\min_x f$. Then $f'(x) = 2\kappa(\varphi(x) - \kappa)/x^3$ with $\varphi(x) = x\sin x - x^2\cos x$, which increases on $(0, \pi)$ and has $\varphi(\pi/2) = \pi/2 > \kappa$. So f has one minimum on $(0, \pi)$, at the $x^* \in (0, \pi/2)$ with $\varphi(x^*) = \kappa$. Elsewhere, f is larger: $f \geq 1$ on $[\pi, 2\pi]$, where $\sin x \leq 0$, and for $x \geq 2\pi$, $f \geq (1 - \kappa/x)^2 > (1 - 2\kappa/\pi)^2 = f(\pi/2)$. This gives (9). As κ grows, $f(x^*)$ falls, since its derivative is $\partial f/\partial\kappa = -2\cos x^* < 0$ (envelope theorem). For the crossings, the output obeys $\dot{y}(t) = (\kappa/L_d)\, e(t - L_d)$, so the error obeys $\dot{e}(t) = -(\kappa/L_d)\, e(t - L_d)$ with $e = 1$ on $[0, L_d]$. It stays positive if and only if its characteristic equation $s + (\kappa/L_d)\mathrm{e}^{-sL_d} = 0$ has a real root; with $w = sL_d$, this reads $-w\mathrm{e}^w = \kappa$, and $-w\mathrm{e}^w$ has the maximum $1/\mathrm{e}$ over real w. Otherwise all solutions oscillate [19]. □

At $\delta = 0.02$, the counts of the loop (8) therefore rise in steps as M_s grows (Fig. 2(a)). With n crossings larger than δ, the computed N_0 and N_1 lie within a quarter turn above $n/2$, and N_2 is $n/2 - 1/4$ within 0.01 (on this ideal loop, $\dot{e}$ jumps at $t = L_d$, so Γ_2 starts with a quarter turn clockwise). The published limits lie between the steps: $\bar{N}_0 = 1/2$, $\bar{N}_1 = 3/4$ and $\bar{N}_2 = 1$ allow 0, 1 and 2 crossings above δ, that is, $M_s \leq 1.534$, 1.842 and 2.210, the values at which the next extremum of e reaches δ. Each band allows one crossing more than the band below, which is why the limits grow; Γ_0 binds first, so on these loops SPIN stops at $M_s \leq 1.534$.

On general loops, this guarantee is lost. The counts leave the steps: in the 3800 count decisions of certified SPIN in Section 5, N_0 and N_2 lay between $n/2 + 1/4$ and $(n+1)/2$ in 23 and 69% of them. And crossings do not bound M_s: the loop of Fig. 1, with a gain margin of 1.624, has $M_s = 2.62$, yet its error never crosses the setpoint by more than 2%, while a controller zero can cause crossings at a low M_s. The counts can shape a tuning, but only a certificate can certify it.

4.3 From certified to safe SPIN

We add the certificates to SPIN in two steps. The first checks each tuning after it is applied, the second before.

Certified SPIN accepts a tuning that the counts accept only if the upper bound $\overline{M}$ *of Theorem 1, computed from the same record, is at most* M^*. In Algorithm 1, it replaces line 7 by the bold lines 7a–7c: the counts shape the tuning, and the certificate decides whether it is accepted. A rejected tuning is lowered by a band move of line 10 that the prediction of Theorem 3 expects to lower M_s, at most 10 in a row, or else by a uniform back-off.

Corollary 1 (Certified SPIN) *Under the conditions of Theorem 2, the result of certified SPIN has* $M_s \leq M^*$, *whatever the limits* $\bar{N}_k$. *If no gain reaches the gain box, the test never traps SPIN: every rejection is followed by an accepted tuning after finitely many moves.*

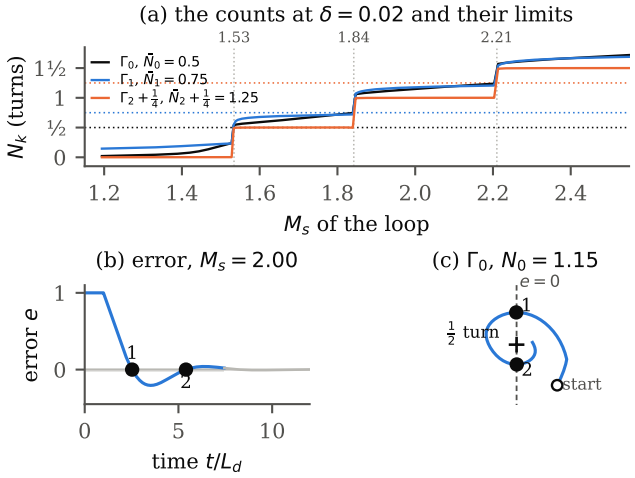


Figure 2: SPIN's counts on the lag-cancelled loop (8). (a) Each count steps over its published limit (dotted) at $M_s = 1.53$, 1.84, 2.21 (N_2 and its limit shifted up a quarter turn). (b) A record with $M_s = 2.00$ and two crossings. (c) Its portrait Γ_0 turns by exactly half a turn between them (Lemma 2); the radius is compressed for display.

Proof: The result is an accepted tuning, so Theorem 2(i) applies. After at most 10 band moves, line 7c backs off uniformly, which ends in an acceptance within j^* steps by Theorem 2(iii), unless line 3 or 10 moves the gains first. □

Certified SPIN ends safe, but it learns that a tuning is fragile only by applying it. *Safe SPIN certifies every move before applying it*, by Theorem 3. It proposes moves only from an accepted tuning, one that passes line 7b on its own record. It computes the move of Algorithm 1 (line 7 or 10) and applies it only if (5) holds and the bound (6) is at most M^*; otherwise it tries the band moves of line 10 for $k = 2, 1, 0$ and then a uniform back-off, and applies the first one so certified; if there is none, it keeps the current tuning. Before its first accepted tuning, it backs off uniformly; if a tuning certified in advance fails line 7b, it returns to the last accepted tuning.

Corollary 2 (Safe SPIN) *Under the conditions of Theorem 2, from the first accepted tuning on, every tuning that safe SPIN applies stabilizes the plant and has $M_s \le M^*$, whatever the limits $\bar{N}_k$.*

Proof: Moves are proposed only from accepted tunings, so each applied tuning is certified by Theorem 3 from the record of an accepted tuning, or is accepted itself (Theorem 2(i)). □

5 Simulations

Setup Each variant runs 200 iterations of Algorithm 1 with its defaults. The derivative has a first-order filter with the time constant $T_f = T_d/10$, where $T_d = K_d/K_p$. Since the gain box ties the result to the start, every loop

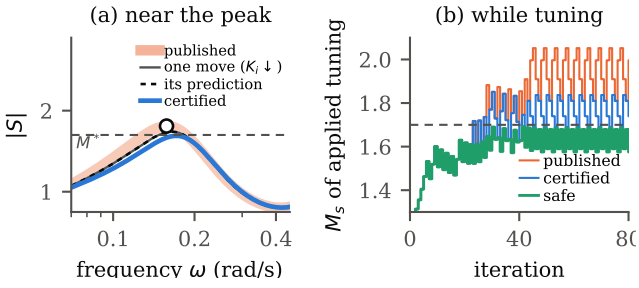


Figure 3: P_3 with PID. (a) Published tuning ($\overline{M} = 1.81$, ○), one band move lowering K_i ($M_s = 1.740$, as predicted by (4)), certified tuning (1.690). (b) M_s of the tuning applied in each iteration.

Table 1: Final M_s (most aggressive tuning accepted in the last 60 iterations; bold: $> M^* = 1.7$) from the cautious (c) and bold (b) start. Bottom: totals over the 19 loops, 3800 iterations per start.

		published		certified		safe	
		c	b	c	b	c	b
P_1	PI	1.46	1.46	1.46	1.46	1.53	1.53
	PID	1.66	**2.03**	1.66	1.62	1.68	1.68
P_2	PI	1.64	1.67	1.64	1.67	1.67	1.67
	PID	1.60	**1.78**	1.68	1.66	1.68	1.69
P_3	PI	1.67	1.63	1.67	1.63	1.65	1.58
	PID	**1.81**	**1.74**	1.69	1.66	1.68	1.69
P_4	PI	1.55	1.60	1.55	1.60	1.56	1.63
	PID	1.61	1.55	1.61	1.55	1.66	1.63
delay-dom.	PI	1.68	1.63	1.68	1.63	1.68	1.67
lag-dom.	PI	**2.01**	1.69	1.67	1.69	1.67	1.69
2 lags	PI	1.44	1.47	1.44	1.47	1.49	1.66
fast	PID	**1.89**	**1.98**	1.63	1.69	1.47	1.50
loops ending $> M^*$		9	7	0	0	0	0
iterations applying $> M^*$		2146	2225	1623	1775	0	0
load-step IAE, mean		–	–	+9%	+8%	+15%	+22%

runs from two starts: SIMC [2] on a first-order-plus-dead-time model obtained by the half rule, with the closed-loop time constant three times the model dead time (cautious start), and the same tuning with all gains doubled (bold start). The 19 test loops are (1) the four plants of [7], $P_1 = e^{-s}/((10s+1)(s+1)^3)$, $P_2 = 1.25e^{-8s}/(5s+1)^4$, $P_3 = e^{-10s}/((2s+1)(s+1)^3)$ and $P_4 = e^{-4s}/(8s+1)^6$, with PI and with PID; (2) four plants that place the oscillation in a chosen band: the delay-dominant $e^{-5s}/(s+1)$ and the lag-dominant $e^{-s}/(10s+1)$ and $e^{-0.5s}/((10s+1)(s+1))$ with PI, and the fast $e^{-0.1s}/((s+1)(0.5s+1))$ with PID; (3) the family $e^{-L_d s}/((s+1)(0.5s+1)(0.25s+1))$ with PID and $L_d \in \{0.25, 0.5, 1, \ldots, 16\}$. P, I and D act on the error. Let T_a be the dead time plus the lag time constants. As in a real controller, T_s stays fixed: $T_s = \min(T_a/100, T_f/20)$ for the PID start tuning and $T_a/1000$ for PI. The lags are discretized exactly and the dead time is rounded to whole samples. Each record covers $15\,T_a$. All values of M_s are those of this digital loop, computed from its exact L. Performance is measured

by the load-step IAE, the integral of $|y|$ after a unit step disturbance at the plant input. Because SPIN ends in a cycle between a few tunings, we report the most aggressive tuning accepted in the last 60 iterations.

The certificate (Section 2) For all 276 tunings accepted in these runs, a four times longer record gave a true remaining variation $\sum_{n>N}|a_n| \le 2.7\cdot 10^{-5}$, so Assumption 1 held with $\tau = 10^{-3}$, and every M_s lay between its bounds, with $\overline{M} - M_s \le 0.003 \approx \eta + \tau$.

Ending below M^* (Corollary 1, Table 1) Published SPIN ended above $M^* = 1.7$ on 9 of the 19 loops from the cautious start and on 7 from the bold start (up to $M_s = 2.07$); certified SPIN on none (at most 1.69), as Theorem 2(i) guarantees, and every loop kept accepting tunings. On P_3 with PID, the predicted band move matched the true M_s (Fig. 3(a)). The load-step IAE rose by 9% and 8% on average.

Staying below M^* (Corollary 2, Table 1, Fig. 3(b)) In the 7600 iterations of the 38 runs, published SPIN applied a tuning with $M_s > M^*$ in 4371 (up to $M_s = 2.38$), certified SPIN, which must apply a tuning to reject it, in 3398 (up to 1.98), and safe SPIN in none: all its tunings were stable, with $M_s \le 1.696$. Every start tuning was accepted, so Corollary 2 covered every iteration. Safe SPIN also ended below M^*, with the same median load-step IAE as certified SPIN; its mean is higher because on the fast PID loop it settled at $M_s \approx 1.5$ with twice the IAE. On 456 random moves (gains scaled by up to 8) from accepted tunings, (5) held for 339, all stable, and failed for 117, all unstable; (6) exceeded M_s by a median of 0.011.

Limit values under the certificate With every limit $\bar{N}_k = 1/2$, certified SPIN still ended below M^*, but 2 and 3 loops (cautious, bold start) accepted no tuning, the median load-step IAE rose by 11 and 19%, and tunings above M^* were still applied in 1310 and 1420 iterations. The limits set performance, not robustness; only certifying in advance keeps the tuning safe.

6 Conclusion

The setpoint step that a model-free tuner applies anyway certifies the robustness of the loop it tests: one finite record gives two-sided bounds on M_s without a model or a dedicated experiment. The same record certifies the stability and M_s of the next tuning before it is applied. With these certificates, any iterative tuner accepts only tunings with $M_s \le M^*$, a uniform back-off always restores acceptance, and no applied tuning exceeds M^*. For SPIN, the turn counts bound M_s only on special loops; they keep shaping the tuning, and the certificates decide what is accepted and applied. Checking a tuning after applying it makes the result safe; checking it before makes the whole tuning safe. The theorems assume a linear loop and a long enough record; noise and real loops are left to a follow-up. Scripts: `https://github.com/ottapav/robopid-simulator`.

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